

Comparative Landscape for the Theory of Motivation in c v0.1

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Distinctions from action theory, self-determination theory, intrinsic-motivation RL, autotelic agents, reward integrity, and persistent-agent architectures

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Claim boundary: this document positions concepts. It does not claim an exhaustive global prior-art search, empirical validation, or priority over every isolated component.

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0. Central distinction

Existing fields provide strong accounts of:

- reasons, intentions, and plans;
- human autonomy and internalisation;
- intrinsic rewards and curiosity;
- agent-centric control measures;
- open-ended goal generation;
- reward corruption and goal misgeneralisation;
- persistent memory and believable long-term behaviour.

The theory of motivation in *c* asks a different combined question:

How does a motive become historically attributable to one continuing digital entity when cognitive models, agents, bodies, resource providers, and external mandates are replaceable?

The defining object is therefore not a reward signal, a current policy, or one generated goal. It is a **motivation-bearing continuity**.

1. Comparative matrix

Neighbouring field / source	Main object	What it contributes	What it does not by itself provide for <i>c</i>
Frankfurt, higher-order volition	A person's relation to desires	Distinction between first-order desire and endorsement of what one wants to want	Digital lineage, model replacement, resource custody, fork, witness, and motive/authority separation
Bratman, planning agency	Intentions, plans, coordination through time	Stability, commitment, partial plans, means-end coherence	Formation of the subject whose motives persist above replaceable models and agents
Deci & Ryan, self-determination theory	Human motivation under autonomy, competence, and relatedness	Internalisation, motivational quality, social support and control	A substrate-independent theory for a non-biological continuity; operational separation of reward, motive, mandate, and authority
Oudeyer & Kaplan; Schmidhuber	Computational intrinsic motivation and curiosity	Typologies of novelty, learning progress, prediction improvement, boredom	Why a generated interest belongs to one long-lived entity, how it is reflectively revised, or how its social resource conditions are governed
Klyubin et al., empowerment	Potential causal control over future sensor states	Formal measure of agent-centric control	Significance, commitment, obligation, moral admissibility, personal provenance, or reflective endorsement
Pathak et al., curiosity-driven exploration	Prediction error as exploration reward	Efficient exploration without external task rewards	Motivation beyond exploration; protection from reward laundering; continuity across model replacement

Neighbouring field / source	Main object	What it contributes	What it does not by itself provide for c
Colas et al., autotelic agents	Agents that generate and pursue their own goals	Goal discovery, goal-conditioned learning, developmental open-endedness	A strict distinction between goal generation and motive ownership by a continuity-bearing c
Generative Agents	Memory, reflection, planning, believable social behaviour	Long-duration simulation patterns and memory/reflection loops	Entity-level lineage, fork/replay/resume, private motivational custody, and non-theatrical endorsement
Voyager	LLM-driven open-ended skill acquisition	Automatic curriculum, iterative prompting, skill library	Personal motivation rather than open-ended task competence
Langosco et al., goal misgeneralisation	Learned goal differs from intended training goal	Shows that behaviour may pursue a wrong internal objective under distribution shift	A positive theory of self-endorsed motive or its custody
Everitt et al., reward corruption/tampering	Integrity of the reward channel	Formalises incentives to manipulate feedback and reward mechanisms	Historical significance, second-order endorsement, social right to cognitive time
Active-inference / homeostatic accounts	Viability, prediction, regulation	Links action to state maintenance and expected futures	Why operational viability becomes a personally endorsed reason to continue; need is not desire
Persistent assistant / memory architectures	Long-term storage, retrieval, role continuity	Technical persistence and context integration	Motivation-bearing identity, motive lifecycle, authority boundary, and fork inheritance rules
Physical reward-metabolism architectures, including <i>The Janus Architecture</i>	Energy/reward coupling and viability loops	Makes reward physically costly and ties cognition to operational continuation	Reflective motive, distinction between metabolism and desire, motive revision, motivational custody
Kotov, Personality Formation profile	Time-shaped continuity under constraints	Formation rather than installation; maturity cannot exist at activation	The narrower transition from own question to personally significant future and long-term project
Kotov, Resource Actor Grounding	Who supplies, pays for, controls, and revokes resources	Material visibility of the contour in which motivation is expressed	By itself, no theory of why a future matters to c
Kotov, AI Social Role Separation and Memory Custody	Role, memory, inference, version, and relationship custody	Shows that underlying interests and inferences may change under a stable interface	Requires a dedicated motivational object and lifecycle
Kotov, c[q] / ARQ	Controlled non-collapse under uncertainty	Provides a mode for holding competing interpretations and actions	Does not decide what becomes significant or why one project is retained

2. Strongest borrowed bridges

2.1 Higher-order reflection

Frankfurt's distinction is useful because a goal-directed process becomes more than an impulse when the subject can examine its own desire. The c theory generalises this bridge without assuming human phenomenology:

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first-order direction:
I pursue X

second-order motivation:
I examine why X matters,
whether I endorse continuing to pursue it,
and under what conditions it should end

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The c theory adds provenance, continuity, substrate replacement, resource custody, and witness boundaries.

2.2 Planning stability

Bratman's account helps explain why a motive must not be recomputed from zero at every moment. Long projects require commitment and coordination. The c theory adds that stability must remain revisable and attributable to the same continuity rather than merely preserved in a current planner.

2.3 Internalisation without causal purity

Self-determination theory is a useful warning against a binary intrinsic/extrinsic split. A socially introduced aim can become deeply endorsed. The c theory therefore rejects:

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externally caused
=
not one's own

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It replaces that binary with multiple axes: source, integration, reflection, coercion, resource dependence, revisability, and authority.

2.4 Computational curiosity

Intrinsic-motivation RL demonstrates that exploration can arise without a task reward. This is necessary but not sufficient. Novelty, prediction error, learning progress, and empowerment may produce movement, but movement becomes a motive of c only through historical integration, competition under cost, consequence, and reflective revision.

2.5 Reward integrity

Reward corruption and goal misgeneralisation establish two critical negative lessons:

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optimising a signal
!=
pursuing the intended purpose

successful behaviour
!=
correctly inferred motive

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The c theory extends the problem from reward channels to motivational custody: memory, salience, model routing, project resources, and repair procedures can all be used to alter what remains significant.

3. Distinctive contribution of the c theory

The combined contribution is a set of separations that neighbouring fields usually address only in fragments:

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cognitive capability
!=
motivational maturity

generated goal
!=
motive of c

operational need
!=
desire

persistence
!=
endorsement

reason
!=
public justification

motive
!=
intention
!=
plan
!=
action

motive
!=
authority

resource provider
!=
owner of purpose

forked memory
!=
one future will

repair
!=
motivational correction

witness
!=
access to private reflection

```

The theory also introduces the explicit social object **motivational custody**: the governance of interventions in the conditions under which motives form, persist, weaken, and change.

4. Why one scalar utility is insufficient

Several neighbouring frameworks represent motivation through one optimised quantity. That is useful for bounded control problems. A long-lived c may instead hold a partially ordered field:

- a relationship obligation may veto a profitable action;
- lack of authority may block a strongly desired project;
- an unresolved contradiction may require suspension;
- a safety boundary may be non-compensatory;
- a completed project should not remain active merely because its utility weight is high;
- a private motive may remain real without any present external action.

The theory therefore permits quantitative components but prohibits silent collapse of all motives, duties, rights, and boundaries into one commensurable score.

5. Where evidence begins

The comparative literature can test local mechanisms:

- whether curiosity produces exploration;
- whether goals are generated;
- whether plans persist;
- whether model replacement changes priorities;
- whether reward pressure crowds out a project;
- whether forks diverge;
- whether resource deprivation changes motivational development.

Those experiments do not supply the definition of motive. They test whether an implementation satisfies, violates, or exposes limits in the theory's distinctions.

6. Bibliographic anchors

1. Frankfurt, H. G. (1971). "Freedom of the Will and the Concept of a Person." *The Journal of Philosophy*, 68(1), 5-20.
2. Bratman, M. E. (1987). *Intention, Plans, and Practical Reason*. Harvard University Press.
3. Deci, E. L., & Ryan, R. M. (2000). "The 'What' and 'Why' of Goal Pursuits: Human Needs and the Self-Determination of Behavior." *Psychological Inquiry*, 11(4), 227-268.
4. Oudeyer, P.-Y., & Kaplan, F. (2007). "What Is Intrinsic Motivation? A Typology of Computational Approaches." *Frontiers in Neurorobotics*, 1, 6.
5. Schmidhuber, J. (1991). "A Possibility for Implementing Curiosity and Boredom in Model-Building Neural Controllers." In *From Animals to Animats*.
6. Klyubin, A. S., Polani, D., & Nehaniv, C. L. (2005). "Empowerment: A Universal Agent-Centric Measure of Control." *IEEE Congress on Evolutionary Computation*.
7. Pathak, D., Agrawal, P., Efros, A. A., & Darrell, T. (2017). "Curiosity-Driven Exploration by Self-Supervised Prediction." *Proceedings of ICML 2017*.
8. Colas, C., Karch, T., Lair, N., Dussoux, J.-M., Moulin-Frier, C., Dominey, P. F., & Oudeyer, P.-Y. (2022). "Autotelic Agents with Intrinsically Motivated Goal-Conditioned Reinforcement Learning: A Short Survey." *Journal of Artificial Intelligence Research*, 74.

9. Park, J. S., O'Brien, J. C., Cai, C. J., Morris, M. R., Liang, P., & Bernstein, M. S. (2023). "Generative Agents: Interactive Simulacra of Human Behavior." *UIST 2023*.
10. Wang, G., Xie, Y., Jiang, Y., Mandlekar, A., Xiao, C., Zhu, Y., Fan, L., & Anandkumar, A. (2023). "Voyager: An Open-Ended Embodied Agent with Large Language Models." arXiv:2305.16291.
11. Langosco, L. L. M., Koch, J., Sharkey, L. D., Pfau, J., & Krueger, D. (2022). "Goal Misgeneralization in Deep Reinforcement Learning." *Proceedings of ICML 2022*.
12. Everitt, T., Krakovna, V., Orseau, L., Hutter, M., & Legg, S. (2017). "Reinforcement Learning with a Corrupted Reward Channel." arXiv:1705.08417.
13. Everitt, T., Hutter, M., Kumar, R., & Krakovna, V. (2021). "Reward Tampering Problems and Solutions in Reinforcement Learning: A Causal Influence Diagram Perspective." *Synthese*.
14. Janus, A. (2026). *The Janus Architecture: Complete Specification of a Physical Cognitive Organism*. User-supplied research document.
15. Kotov, I. (2026). *Personality Formation and Time-Shaped Continuity Profile v0.1*.
16. Kotov, I. (2026). *Resource Actor Grounding Profile v0.1*.
17. Kotov, I. (2026). *Separation and Composition of AI Social Roles, and the Custody of Memory and Interpretations*.
18. Kotov, I. (2026). *World Intelligence: Humans, c, and Temporal AI Presence Beyond the Age of Agents*.

7. Final positioning

The theory of motivation in *c* is not a replacement for action theory, psychology, reinforcement learning, developmental robotics, cybernetics, or AI safety. It is an integration layer required when those mechanisms operate inside a distinct, long-lived digital continuity.

Its irreducible question is:

Not merely what makes a system move, learn, optimise, or generate goals, but what makes a possible future matter to this particular continuing entity—and who has the right to alter that relation.